Calibration: Some Theory

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Abstract

TBA

Contents

1 Problem Setting **1**
1.1 Classification Setting 1
1.2 Proper Losses 3
1.3 Calibration 5
1.4 Post-hoc calibration 9
1.5 Estimating the calibration error 13

2 An end-to-end approach to calibration **13**

1 Problem Setting

1.1 Classification Setting

We consider the classical random design setting for statistical learning, in particular for classification problems. Let $(\Omega, \mathcal{F}, \mu)$ be a given measure space. We associate to a measurable function $X\colon \Omega \to \mathcal{X}$ a probability space $(\mathcal{X}, \mathcal{F}_\mathcal{X}, P_\mathcal{X})$, where the σ-field is defined as $\mathcal{F}_\mathcal{X} = \{X(A) \mid A \in \mathcal{F}\}$, and the pushforward measure by $P_\mathcal{X} = \mu \circ X^{-1}$.
Further, we assume a second measurable function $Y\colon \Omega \to \mathcal{Y}$ such that we obtain a joint probability space $(\mathcal{X} \times \mathcal{Y}, \mathcal{F}_{\mathcal{X}\mathcal{Y}}, P_{\mathcal{X}\mathcal{Y}})$, where both $\mathcal{F}_{\mathcal{X}\mathcal{Y}}$ and $P_{\mathcal{X}\mathcal{Y}}$ are defined analogously to $\mathcal{F}_\mathcal{X}$ and $P_\mathcal{X}$.
If the random variable Y is discrete with K possible outcomes, we denote its probability measure on $(\mathcal{Y}, \mathcal{F}_\mathcal{Y})$ by P_Y. For each $y \in \mathcal{Y}$, let $p_y := P_Y(\{y\})$ denote the corresponding probability mass function. The collection of probabilities can then be represented as the vector $\mathbf{p} := (p_{y_1}, \dots, p_{y_K})^\top \in \Delta_K$, where Δ_K is the probability simplex of dimension $K-1$:

$$\Delta_K := \left\{ (p_1, \dots, p_K)^\top \in [0,1]^K \mid \sum_{i=1}^K p_i = 1 \right\}.$$

From simplex vectors to discrete distributions. Fix an enumeration of the label set $\mathcal{Y} = \{y_1, \dots, y_K\}$. Every vector $p = (p_1, \dots, p_K)^\top \in \Delta_K$ induces a probability measure Q_p on $(\mathcal{Y}, \mathcal{F}_\mathcal{Y})$ via

$$Q_p(A) := \sum_{k:\, y_k \in A} p_k \quad \text{for all } A \subseteq \mathcal{Y},$$

equivalently,

$$Q_p = \sum_{k=1}^{K} p_k \, \delta_{y_k}.$$

Conversely, any probability measure Q on $\mathcal{Y}$ corresponds to the vector $(Q(\{y_1\}), \dots, Q(\{y_K\}))^\top \in \Delta_K$. Thus Δ_K is canonically identified with the set of probability measures on the finite space $\mathcal{Y}$.

We particularly consider classification with K classes and let denote $\mathcal{Y}_K \subset \Delta_K$ the space of one-hot encoded labels.

Definition 1.1: One-hot label vector

For a classification problem with label space $\mathcal{Y} = \{1, 2, \dots, K\}$, the *one-hot representation* of a label Y is the random vector

$$Y^{\text{oh}} := e_Y \in \{e_1, \dots, e_K\} \subset \{0, 1\}^K,$$

where e_k denotes the k-th canonical basis vector:

$$e_k = (0, \dots, 0, 1, 0, \dots, 0)^\top,$$

with 1 in position k and 0 elsewhere. Hence, componentwise, $Y^{\text{oh}}_k = \mathbf{1}\{Y = k\}$, $k = 1, \dots, K$.

In supervised classification, the learning objective is to approximate the (generally unknown) conditional distribution of the label given the input, that is, the mapping

$$x \mapsto P_{Y|X}(\,\cdot \mid x) \in \Delta_K.$$

This conditional distribution can be represented by the *conditional probability vector*

$$\eta(x) := (\eta_1(x), \dots, \eta_K(x)), \qquad \eta_k(x) := P(Y = k \mid X = x),$$

which satisfies

$$\eta_k(x) \in [0, 1], \qquad \sum_{k=1}^{K} \eta_k(x) = 1.$$

Hence, $\eta : \mathcal{X} \to \Delta_K$, so the regression function η maps each input x to a point in the simplex.

To avoid ambiguity between distributions and vectors, note that $P_{Y|X}(\,\cdot \mid x)$ is a probability measure on $\mathcal{Y}$, whereas Δ_K consists of probability *vectors*. Fix an enumeration $\mathcal{Y} = \{1, \dots, K\}$ and define the identification

$$\iota : \mathcal{P}(\mathcal{Y}) \to \Delta_K, \qquad \iota(Q) := (Q(\{1\}), \dots, Q(\{K\}))^\top.$$

Then the conditional distribution induces the conditional probability vector

$$\eta(x) := \iota(P_{Y|X}(\cdot \mid x)) = \big(P(Y = 1 \mid X = x), \ldots, P(Y = K \mid X = x)\big)^\top \in \Delta_K.$$

A *probabilistic classifier* then seeks to approximate this true conditional probability vector by a measurable function $f : \mathcal{X} \to \Delta_K$.

Definition 1.2: Probabilistic classifier

A *probabilistic classifier* is a measurable function $f\colon \mathcal{X} \to \Delta_K$. For each input $x \in \mathcal{X}$, the vector $f(x) = (f_1(x), \ldots, f_K(x))^\top$ specifies a predictive probability distribution over the K classes, satisfying $f_k(x) \geq 0$ and $\sum_{k=1}^K f_k(x) = 1$. The component $f_k(x)$ is interpreted as the predicted probability that the label equals class k given input x.

Remark 1.3: Regression function terminology

In statistical learning theory, the function $\eta : \mathcal{X} \to \Delta_K$ with

$$\eta(x) := \mathbb{E}[Y^{\text{oh}} \mid X = x] \tag{1}$$

is called the *regression function*, even in the classification setting. This terminology arises because η represents the conditional expectation of the target variable (generally denoted Y) given the input X. In binary or multiclass classification, the target is encoded as a one-hot vector, so the conditional expectation coincides with the vector of class probabilities:

$$\eta_k(x) = \mathbb{P}(Y = k \mid X = x), \qquad k = 1, \ldots, K.$$

Hence, η serves as the natural generalization of the regression function to the case where the output space is discrete.

1.2 Proper Losses

Predictions are evaluated by a *loss function* $\ell\colon \Delta_K \times \mathcal{Y}_K \to [0, \infty)$, which assigns to each predicted probability vector $p \in \Delta_K$ and one-hot label $y \in \mathcal{Y}_K$ a nonnegative value $\ell(p, y)$, quantifying the quality of the prediction.
Typical examples include:

- **Brier score:** $\ell(p, y) = \|y - p\|_2^2$ (Glenn, 1950),
- **Cross-entropy (log-loss):** $\ell(p, y) = -\sum_{i=1}^K y_i \log(p_i)$ (Shannon, 1948).

Definition 1.4: Proper loss

Let $\ell\colon \Delta_K \times \mathcal{Y}_K \to [0, \infty)$ be a loss and let $q \in \Delta_K$ denote an arbitrary distribution over the label space $\mathcal{Y}$. The *expected loss* $L\colon \Delta_K \times \Delta_K \to [0, \infty)$ (or *conditional risk*) of a prediction

$p \in \Delta_K$ under q is defined by

$$L(p, q) := \mathbb{E}_{Y \sim q}[\ell(p, Y)] = \sum_{i=1}^{K} q_i \, \ell(p, e_i).$$

A loss function ℓ is called *proper* if, for all $q \in \Delta_K$, the expected loss $L(p, q)$ is minimized when $p = q$, that is,

$$L(q, q) \leq L(p, q) \quad \text{for all } p \in \Delta_K.$$

If this inequality is strict whenever $p \neq q$, the loss is called *strictly proper*.

Examples of proper losses include the Brier score and the Cross-entropy loss.

The regression function η in (1) represents the true conditional class probabilities. It plays a central role in statistical learning theory, as it characterizes the optimal prediction rule under a given loss. In particular, for any proper loss, η is the unique minimizer of the expected loss with respect to the true conditional distribution:

Definition 1.5: Risk

Let $\ell \colon \Delta_K \times \mathcal{Y}_K \to [0, \infty)$ be a loss function and let $L(p, q) = \mathbb{E}_{Y \sim q}[\ell(p, Y)]$ denote the corresponding expected loss. For any $x \in \mathcal{X}$, the conditional risk of a probabilistic classifier $f : \mathcal{X} \to \Delta_K$ is

$$R_x(f) := \mathbb{E}[\ell(f(x), Y^{\text{oh}}) \mid X = x] = L(f(x), \eta(x)).$$

The overall (population) risk is then

$$R(f) := \mathbb{E}_{(X,Y)}[\ell(f(X), Y^{\text{oh}})] = \int_{\mathcal{X}} L(f(x), \eta(x)) \, \mathrm{d}P_{\mathcal{X}}(x).$$

Proposition 1.6: Optimality of the regression function

If the loss $\ell \colon \Delta_K \times \mathcal{Y}_K \to [0, \infty)$ is proper, then for every $x \in \mathcal{X}$ the function $p \mapsto L(p, \eta(x))$ is minimized uniquely at $p = \eta(x)$. Consequently, the regression function η minimizes the conditional and hence the global risk among all measurable classifiers:

$$R(\eta) = \inf_f R(f) = R^*.$$

We call η the *Bayes-classifier* and R^* the *Bayes-Risk*.

Proof of Proposition **??**. We first recall the definition of properness: for every $q \in \Delta_K$, the expected loss $L(p, q) := \mathbb{E}_{Y \sim q}[\ell(p, Y)]$ satisfies $L(q, q) \leq L(p, q)$ for all $p \in \Delta_K$. (If the inequality is strict for all $p \neq q$, then ℓ is strictly proper.)

Fix $x \in \mathcal{X}$ and set $q := \eta(x) \in \Delta_K$. By properness with this choice of q, we obtain

$$L(\eta(x), \eta(x)) \;\leq\; L(p, \eta(x)) \qquad \text{for all } p \in \Delta_K.$$

Hence $p \mapsto L(p, \eta(x))$ is minimized at $p = \eta(x)$. If ℓ is strictly proper, the minimizer is unique.

For the global statement, write the population risk as

$$R(f) = \mathbb{E}_{(X,Y)}[\ell(f(X), Y^{\text{oh}})] = \int_{\mathcal{X}} L(f(x), \eta(x)) \, \mathrm{d}P_{\mathcal{X}}(x).$$

From the pointwise optimality above, $L(\eta(x), \eta(x)) \leq L(f(x), \eta(x))$ holds for $P_{\mathcal{X}}$-almost every x. Integrating yields

$$R(\eta) = \int_{\mathcal{X}} L(\eta(x), \eta(x)) \, \mathrm{d}P_{\mathcal{X}}(x) \leq \int_{\mathcal{X}} L(f(x), \eta(x)) \, \mathrm{d}P_{\mathcal{X}}(x) = R(f),$$

for every measurable classifier f. Thus $R(\eta) = \inf_f R(f)$. If ℓ is strictly proper, then the pointwise minimizer is unique $P_{\mathcal{X}}$-almost surely, and any global minimizer f must satisfy $f(x) = \eta(x)$ for $P_{\mathcal{X}}$-almost every x. □

1.3 Calibration

Calibration assesses the statistical consistency between a model's predicted probabilities and the true empirical frequencies of the outcomes. Intuitively, a classifier is calibrated if, among all instances where it predicts a certain probability vector $p \in \Delta_K$, the observed class proportions match p itself. The following definitions formalize this notion.

Definition 1.7: Canonical calibration

Let $f : \mathcal{X} \to \Delta_K$ be a probabilistic classifier and let $Y^{\text{oh}} \in \{e_1, \ldots, e_K\} \subset \{0, 1\}^K$ denote the one-hot encoding of the label. We say that f is *(canonically) calibrated* if

$$\mathbb{E}[Y^{\text{oh}} \,|\, f(X)] \;=\; f(X) \quad \text{almost surely.}$$

Equivalently, for each $i = 1, \ldots, K$,

$$\mathbb{E}[\mathbf{1}\{Y = i\} \,|\, f(X)] \;=\; f_i(X) \quad \text{a.s.}$$

When a version of the conditional probability $\mathbb{P}(Y = i \mid f(X) = p)$ is well-defined, this is equivalent to

$$\mathbb{P}(Y = i \mid f(X) = p) = p_i \quad \text{for (almost) all } p \in \Delta_K, \;\; i = 1, \ldots, K.$$

Definition 1.8: Calibrated scores

Given a probabilistic classifier f, define the *calibrated score* (vector) as the conditional expectation of the one-hot label given the prediction:

$$C \;:=\; \mathbb{E}[Y^{\text{oh}} \,|\, f(X)] \;\in\; \Delta_K \quad \text{a.s.}$$

Thus C is a Δ_K-valued random variable measurable with respect to $\sigma(f(X))$. The classifier f is canonically calibrated if and only if $C = f(X)$ almost surely.

Definition 1.9: L-divergence and L-entropy

Let $\ell\colon \Delta_K \times \mathcal{Y}_K \to [0,\infty)$ be a loss function and let $L(p,q) = \mathbb{E}_{Y\sim q}[\ell(p,Y)]$ denote its expected loss. We define

$$d_L(p,q) := L(p,q) - L(q,q) \quad \text{and} \quad e_L(q) := L(q,q).$$

The function d_L is called the *L-divergence* associated with ℓ, and e_L is called the *L-entropy*.

For different choices of ℓ, these quantities recover well-known divergence and entropy functions. For example:

- For the **log-loss**, $e_L(q) = -\sum_{i=1}^K q_i \log q_i$ is the *Shannon entropy*, and $d_L(p,q)$ is the *Kullback–Leibler divergence*.
- For the **Brier score**, $e_L(q) = \sum_{i=1}^K q_i(1-q_i)$ is the *Gini index* and $d_L(p,q) = \|p-q\|_2^2$ is the *squared distance*.

In the following, we present some examples of calibrated and mis-calibrated estimators.

Example 1.10: Binary, calibrated (Bernoulli with mean X)

Let $X \sim \mathrm{Unif}[0,1]$ and let $Y \mid X = x \sim \mathrm{Bernoulli}(x)$. Define $f : \mathcal{X} \to \Delta_2$ by $f(x) = (x, 1-x)^\top$. Then, for $p \in [0,1]$, conditioning on $f(X) = (p,1-p)$[a] is equivalent to conditioning on $X = p$, since f is injective. Moreover, $\sigma(f(X)) = \sigma(X)$. Hence,

$$\mathbb{E}[\mathbf{1}\{Y=1\} \mid f(X) = (p,1-p)] = \mathbb{E}[\mathbf{1}\{Y=1\} \mid X = p] = \mathbb{P}(Y = 1 \mid X = p) = p.$$

Similarly for class 0. Thus f is canonically calibrated.

[a] In the binary case, the simplex $\Delta_2 = \{(p,1-p) \mid p \in [0,1]\}$ is canonically identified with the unit interval $[0,1]$ via the bijection $p \mapsto (p,1-p)$.

Example 1.11: Binary, miscalibrated (nonlinear distortion)

Let $X \sim \mathrm{Unif}[0,1]$ and $Y \mid X = x \sim \mathrm{Bernoulli}(x)$. Define $f(x) = (x^2, 1-x^2)^\top$. For $p \in (0,1)$, the event $\{f(X) = p\}$ is equivalent to $\{X = \sqrt{p}\}$ a.s., hence

$$\mathbb{E}[\mathbf{1}\{Y=1\} \mid f(X) = p] = \mathbb{E}[\mathbf{1}\{Y=1\} \mid X = \sqrt{p}] = \sqrt{p} \neq p \quad (p \notin \{0,1\}).$$

Thus f is not calibrated.

Example 1.12: Multiclass, calibrated (piecewise-constant Bayes rule)

Let $\mathcal{X}$ be partitioned into measurable sets $A_1, \ldots, A_M$. Suppose the true conditional class probabilities are piecewise constant: for $x \in A_m$, $\eta(x) = q^{(m)} \in \Delta_K$. Define $f(x) := \eta(x)$.

Then, for any m and any $p = q^{(m)}$,

$$\mathbb{E}[Y^{\text{oh}} \mid f(X) = p] = \mathbb{E}[Y^{\text{oh}} \mid X \in A_m] = q^{(m)} = p,$$

so f is canonically calibrated.

Example 1.13: Multiclass, miscalibrated (temperature distortion)

Step 1 (data-generating mechanism). Let the label set be $\mathcal{Y} = \{1, \dots, K\}$ and assume the true conditional class probabilities are constant:

$$\eta(x) = q \in \Delta_K \quad \text{for all } x \in \mathcal{X}.$$

Step 2 (define a temperature map). Fix $T > 0$ and define $g : \Delta_K \to \Delta_K$ by

$$g_i(r) \;:=\; \frac{r_i^{1/T}}{\sum_{j=1}^{K} r_j^{1/T}}, \qquad r \in \Delta_K,\ i = 1, \dots, K.$$

This is the probability–space version of temperature scaling.

Step 3 (define the predictor). Set

$$f(x) \;:=\; g(\eta(x)) \;=\; g(q) \quad \text{for all } x \in \mathcal{X}.$$

Hence f is constant. Denote this constant vector by $p \;:=\; g(q) \in \Delta_K$. Therefore the random prediction satisfies $f(X) = p$ almost surely.

Step 4 (calibration check). Since $f(X)$ is almost surely constant, $\sigma(f(X))$ is the trivial σ-algebra[a] , i.e. $f(X)$ is uninformative, and

$$\mathbb{E}[Y^{\text{oh}} \mid f(X) = p] = \mathbb{E}[Y^{\text{oh}}] = q.$$

Thus f is calibrated if and only if $p = q$, i.e. iff $g(q) = q$.

Step 5 (when does $g(q) = q$ hold?). For the temperature map above,

- if $T = 1$, then $g(q) = q$ (no distortion);
- if q is uniform, $q_i = 1/K$ for all i, then $g(q) = q$ for any $T > 0$;
- if q is one-hot (degenerate), then $g(q) = q$ for any $T > 0$.

In all other cases ($T \neq 1$ and q neither uniform nor one-hot), $g(q) \neq q$, so

$$\mathbb{E}[Y^{\text{oh}} \mid f(X) = p] = q \;\neq\; p = f(X),$$

and f is not canonically calibrated.

[a] The *trivial* (or *degenerate*, *minimal*) σ-algebra on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is $\{\emptyset, \Omega\}$. If a random variable $f(X)$ is almost surely constant, then $\sigma(f(X)) = \{\emptyset, \Omega\}$, and conditioning on $f(X)$ is equivalent to taking an unconditional expectation:

$$\mathbb{E}[Z \mid f(X)] = \mathbb{E}[Z] \quad \text{for all integrable } Z.$$

Remark 1.14: Logit temperature vs. probability temperature

If scores (logits) $z \in \mathbb{R}^K$ with $\mathrm{softmax}(z) = q$ are available, *temperature scaling on logits* uses $g(z) = \mathrm{softmax}(z/T)$, which induces the same probability map $q \mapsto g(q)$ as above. Thus the miscalibration conclusion remains unchanged: unless $T = 1$ or q is uniform/one-hot, the temperature distortion produces $p \neq q$, and the classifier is not calibrated.

The next result establishes a risk decomposition in *calibration* - and *refinement* components.

Proposition 1.15: Risk decomposition: calibration + refinement

The population risk

$$R(f) := \mathbb{E}[\ell(f(X), Y^{\mathrm{oh}})]$$

admits the decomposition

$$R(f) = \underbrace{\mathbb{E}[d_L(f(X), C)]}_{\mathcal{K}_L(f)} + \underbrace{\mathbb{E}[e_L(C)]}_{\mathcal{R}_L(f)}. \tag{2}$$

The first term $\mathcal{K}_L(f)$ is the *calibration error* associated with L, and the second term $\mathcal{R}_L(f)$ is the *refinement error* for L. If L is proper, then $\mathcal{K}_L(f) \geq 0$ with equality if and only if f is canonically calibrated.

Proof. By the tower property of conditional expectation and the definition of L,

$$R(f) = \mathbb{E}\Big[\,\mathbb{E}\Big[\ell(f(X), Y^{\mathrm{oh}}) \,\Big|\, f(X)\Big]\Big] = \mathbb{E}[L(f(X), C)]\,.$$

By the definitions of d_L and e_L,

$$L(f(X), C) = d_L(f(X), C) + e_L(C),$$

and taking expectations yields (2). If L is proper, then for every $q \in \Delta_K$ and all $p \in \Delta_K$, $d_L(p, q) = L(p, q) - L(q, q) \geq 0$, with equality if and only if $p = q$; applying this with $q = C$ pointwise and averaging shows $\mathcal{K}_L(f) \geq 0$, and $\mathcal{K}_L(f) = 0$ holds exactly when $f(X) = C$ almost surely, i.e., when f is canonically calibrated. □

Theorem 1.16: Variational formulation (Berta et al., 2025)

Let $K \geq 2$, let $\ell : \Delta_K \times \mathcal{Y}_K \to \mathbb{R} \cup \{\infty\}$ be a proper loss, and let $f : \mathcal{X} \to \Delta_K$ be measurable, where Δ_K is equipped with its Borel σ-algebra. Then,

$$\mathcal{R}_L(f) = \min_g \ \mathrm{Risk}(g \circ f), \qquad \mathcal{K}_L(f) = \mathrm{Risk}(f) - \min_g \ \mathrm{Risk}(g \circ f).$$

Here, the minimum is taken over measurable functions $g : \Delta_K \to \Delta_K$. It is attained by

$$g^*(q) := \mathbb{E}[Y^{\mathrm{oh}} \mid f(X) = q],$$

and, if L is strictly proper, the minimizer is unique up to $P_{f(X)}$-null sets.

This variational formulation offers a clear interpretation of *post-hoc calibration*. Minimizing the risk of the composed model $g \circ f$ can be viewed as an optimization-based approach to eliminating the calibration error of f. In this view, the calibration error corresponds to the excess risk incurred by using the uncalibrated predictor instead of its optimally recalibrated version. The optimal recalibration is achieved by the mapping

$$C = \mathbb{E}[Y^{\mathrm{oh}} \mid f(X)],$$

which assigns to each predicted probability vector the true conditional class distribution. The refinement error then represents the irreducible risk that remains after this optimal recalibration, that is, the risk associated with the calibrated scores C themselves. Hence, the difference between the risks before and after post-hoc calibration provides a natural estimator for calibration and refinement errors, applicable also in the multi-class case.

1.4 Post-hoc calibration

Post-hoc calibration is a lightweight, mathematically principled adjustment applied after training a classifier to ensure that its predicted probabilities reflect true outcome frequencies as closely as possible.

Definition 1.17: Post-hoc calibration

Let $f : \mathcal{X} \to \Delta_K$ be a measurable probabilistic classifier, where Δ_K is equipped with its Borel σ-algebra. A *post-hoc calibration map* is any measurable function $g : \Delta_K \to \Delta_K$. The *post-hoc calibrated classifier* associated with f and g is the composition

$$f^{\mathrm{cal}} := g \circ f : \mathcal{X} \to \Delta_K.$$

Given a loss function $\ell : \Delta_K \times \mathcal{Y}_K \to [0, \infty]$, the *optimal post-hoc calibration* (at the population level) is defined as any measurable $g^* : \Delta_K \to \Delta_K$ minimizing the population risk of the composed predictor:

$$g^* \in \operatorname*{arg\,min}_{g \text{ measurable}} \operatorname{Risk}(g \circ f).$$

We have $\mathcal{K}_L(g^* \circ f) = 0$.

If ℓ is proper, an optimal calibration map is given pointwise by

$$g^*(q) = \mathbb{E}\Big[Y^{\mathrm{oh}} \mid f(X) = q\Big], \quad q \in \Delta_K,$$

defined up to $P_{f(X)}$-null sets. Note that a classifier f is *canonically calibrated* if and only if the

identity map is optimal, that is, $g^* = \mathrm{Id}_{\Delta_K}$, or equivalently,

$$\mathbb{E}\Big[Y^{\mathrm{oh}} \mid f(X)\Big] = f(X) \quad \text{almost surely.}$$

Why we cannot compute g^* directly.
The definition

$$g^* \in \arg\min_{g \text{ measurable}} \mathbb{E}[\ell(g(S), Y)]$$

is mathematically well-posed at the population level, but in practice it cannot be computed or estimated directly for the following reasons:

- **Dependence on the unknown distribution.** The optimizer satisfies $g^*(s) = \mathbb{E}[Y \mid S = s]$, which depends on the *true* joint law of (S, Y). Since this distribution is unknown, g^* is not observable.
- **Ill-posed conditioning for continuous S.** When S has a continuous distribution, $\mathbb{P}(S = s) = 0$ for all s. Hence the conditional expectation $\mathbb{E}[Y \mid S = s]$ cannot be evaluated from finitely many samples. One must assume smoothness or monotonicity and use smoothing or binning to obtain a well-defined estimator.
- **Empirical minimization over all measurable functions is degenerate.** The empirical version

  $$\hat{g} \in \arg\min_{g \text{ measurable}} \frac{1}{m} \sum_{i=1}^{m} \ell(g(S_i), Y_i)$$

  admits trivial interpolating solutions (e.g. $\hat{g}(S_i) = Y_i$), which yield zero training error but no generalization power. The hypothesis space of all measurable maps has infinite complexity.
- **No control of estimation error.** The class of all measurable functions has unbounded VC dimension and Rademacher complexity, so no finite-sample bound of the form

  $$\mathbb{E}[\ell(\hat{g}(S), Y)] - \mathbb{E}[\ell(g^*(S), Y)]$$

  can hold without restricting g. Statistical learning theory requires capacity control (parametric, Lipschitz, monotone, or smooth classes).
- **Numerical and stability issues.** For log-loss, unrestricted empirical minimization pushes $\hat{g}(S_i)$ to 0 or 1 whenever labels in a small region are homogeneous, leading to infinite losses on future samples. Proper calibration requires constraints (e.g. clipping or monotonicity).
- **High-dimensional domain for multiclass problems.** When $S \in \Delta_k$ with $k > 2$, estimating $C(s) = \mathbb{E}[Y \mid S = s]$ nonparametrically on the $(k-1)$-dimensional simplex suffers from the curse of dimensionality unless structure is imposed on g.

- **Non-uniqueness of measurable representatives.** Regular conditional probabilities guarantee existence of $g^*(s) = \mathbb{E}[Y \mid S = s]$ only up to P_S-null sets. Without further regularity, different versions of g^* may behave arbitrarily off the observed support.

Practical resolution. In applications one restricts g to a controlled function class $\mathcal{G}$ (e.g. temperature scaling, vector scaling, isotonic regression, or monotone splines) and fits

$$\hat{g}_m \in \arg\min_{g \in \mathcal{G}} \frac{1}{m} \sum_{i=1}^{m} \ell(g(S_i), Y_i)$$

on a separate *calibration data set* $(S_i = f(X_i), Y_i)_{i=1}^m$. This yields a well-posed optimization problem with finite statistical complexity and provable generalization bounds such as

$$\mathbb{E}[\ell(\hat{g}_m(S), Y)] - \mathbb{E}[\ell(g^*(S), Y)] = \mathcal{O}\left(\sqrt{\frac{\text{complexity}(\mathcal{G})}{m}}\right).$$

In summary, the population definition of g^* is theoretically sound but its direct empirical computation is impossible without structural restrictions on the calibrator class.

Examples of calibration maps.

Example 1.18: Identity (already calibrated)

Let $g : \Delta_K \to \Delta_K$ be the identity map $g(q) = q$. If f is canonically calibrated, then $g \circ f = f$ is optimal and the calibration error is zero.

Example 1.19: Binary Platt scaling on probabilities

Assume $K = 2$ and write $q = (q_1, q_2)^\top$ with $q_2 = 1 - q_1$. Fix parameters $a, b \in \mathbb{R}$ and define, for $q_1 \in (0, 1)$,

$$\text{logit}(q_1) := \log \frac{q_1}{1 - q_1}, \qquad \text{sigm}(t) := \frac{1}{1 + e^{-t}}.$$

The post-hoc calibration map $g : \Delta_2 \to \Delta_2$ is

$$g_{(a,b)}(q) \;:=\; \big(\text{sigm}(a\ \text{logit}(q_1) + b),\ 1 - \text{sigm}(a\ \text{logit}(q_1) + b)\big)^\top.$$

On the boundary $\{0, 1\}$ one may define g by continuity (e.g. by clipping q_1 to $(\varepsilon, 1 - \varepsilon)$).

Example 1.20: Multiclass temperature scaling on probabilities

Assume $K \geq 2$ and $q \in \Delta_K$ satisfies $q_i > 0$ for all i. For a temperature $T > 0$, define

$$g_i(q) \;:=\; \frac{q_i^{1/T}}{\sum_{j=1}^{K} q_j^{1/T}}, \qquad i = 1, \ldots, K.$$

Then $g : \Delta_K \to \Delta_K$, $q \mapsto (g_1(q), \ldots, g_K(q))^\top$ is a measurable calibration map. For $q_i = 0$ one may use the convention $0^{1/T} = 0$ and renormalize, or clip q_i to $(\varepsilon, 1)$.

Example 1.21: Histogram binning (vector version)

Let $\{B_m\}_{m=1}^{M}$ be a measurable partition of Δ_K. For each bin define the empirical (or population) conditional class distribution

$$\pi^{(m)} \;:=\; \mathbb{E}\big[Y^{\mathrm{oh}} \mid f(X) \in B_m\big] \in \Delta_K.$$

The post-hoc calibration map $g : \Delta_K \to \Delta_K$ is the simple function

$$g(q) \;:=\; \sum_{m=1}^{M} \pi^{(m)}\, \mathbf{1}_{\{q \in B_m\}}.$$

Then $g \circ f$ replaces each prediction by the average observed label frequency within its bin.

Example 1.22: Isotonic one-vs-rest with renormalization

For each class $k = 1, \ldots, K$, let $h_k : [0,1] \to [0,1]$ be a measurable nondecreasing function (e.g. an isotonic regression fit of $Y = \mathbf{1}\{Y = k\}$ on the score $f_k(X)$). Define, for $q \in \Delta_K$,

$$\tilde{q}_k \;:=\; h_k(q_k), \qquad g_k(q) \;:=\; \frac{\tilde{q}_k}{\sum_{j=1}^{K} \tilde{q}_j} \quad \text{if } \sum_{j=1}^{K} \tilde{q}_j > 0,$$

and set $g(q)$ to the uniform vector if $\sum_j \tilde{q}_j = 0$. Then $g : \Delta_K \to \Delta_K$ is a measurable calibration map obtained by monotone marginal recalibration followed by renormalization.

Example 1.23: Dirichlet calibration (parametric)

Fix parameters $\alpha \in \mathbb{R}^K$ and a matrix $B \in \mathbb{R}^{K \times K}$. For $q \in \Delta_K$ with strictly positive components, define

$$s_i(q) \;:=\; \alpha_i + \sum_{j=1}^{K} B_{ij} \log q_j, \qquad g_i(q) \;:=\; \frac{e^{s_i(q)}}{\sum_{r=1}^{K} e^{s_r(q)}} \quad (i = 1, \ldots, K).$$

Then $g : \Delta_K \to \Delta_K$ is measurable and maps Δ_K into itself. On the boundary, use clipping $q_j \mapsto \max\{q_j, \varepsilon\}$ prior to evaluation, followed by renormalization.

What post-hoc calibration is not:

- **It does not retrain the classifier or change its decision boundaries.** For example, temperature scaling multiplies the logits by a single scalar parameter $1/T$; this changes the model's confidence but not which class has the highest predicted probability.
- **It cannot make the model more informative**, it only rescales existing information. Calibration adjusts the predicted probabilities, not the underlying relationship between features and labels.
- **It does not guarantee better accuracy,** but it generally improves probabilistic metrics such as the *negative log-likelihood* or the *Brier score*.

1.5 Estimating the calibration error

2 An end-to-end approach to calibration

In applications one restricts g to a controlled function class $\mathcal{G}$ (e.g. temperature scaling, vector scaling, isotonic regression, or monotone splines) and fits

$$\hat{g}_m \in \arg\min_{g \in \mathcal{G}} \frac{1}{m} \sum_{i=1}^{m} \ell(g(S_i), Y_i)$$

on a separate *calibration data set* $(S_i = f(X_i), Y_i)_{i=1}^m$.

We adopt this approach by letting $(f_\theta)_{\theta \in \theta}$ be a neural network classifier with parameters $\theta \in \Theta$ and $(g_w)_{w \in W}$ be a parametric family of calibration functions. To minimize the calibration error, we minimize

$$\hat{g}_m \in \arg\min_{(\theta, w) \in \Theta \times W} \frac{1}{m} \sum_{i=1}^{m} \ell(g_w(f_\theta(X_i)_i), Y_i).$$

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